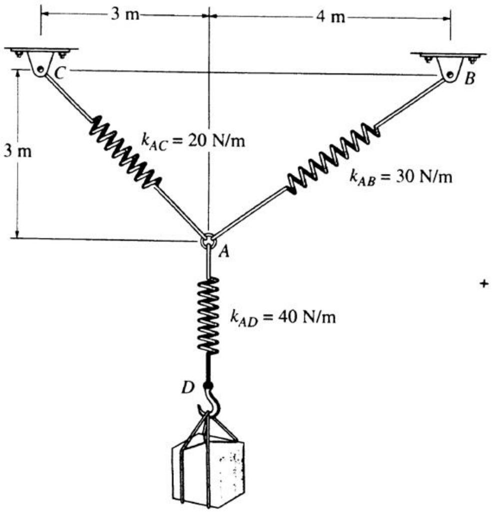
भारतीय प्रौद्योगिकी संस्थान जोधपुर
एनएच 65, सुरपुरा बाईपास रोड, करवड़, राजस्थान 342037

Indian Institute of Technology Jodhpur
NH 65, Surpura Bypass Rd, Karwar, Rajasthan 342037

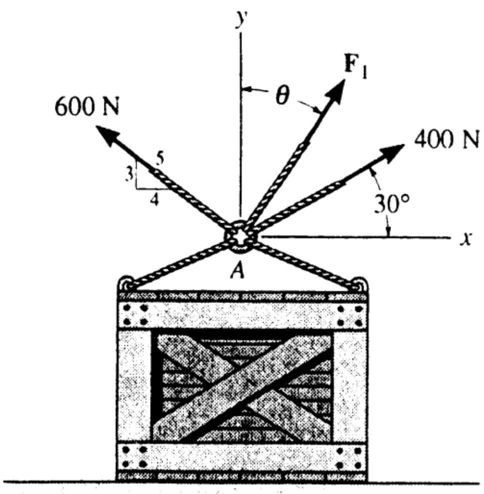
Course: Engineering Mechanics **Code: MEL1010**

Date: 23/08/2024 **Tutorial: 02**

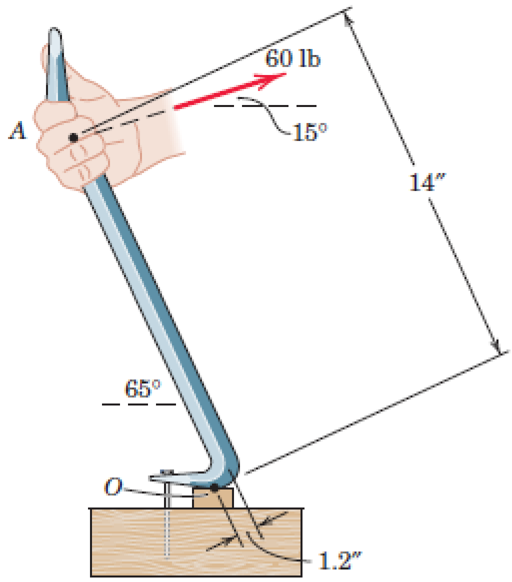
Question 01: Determine the stretch in each spring for equilibrium of 2 kg block. The springs are shown in equilibrium position.



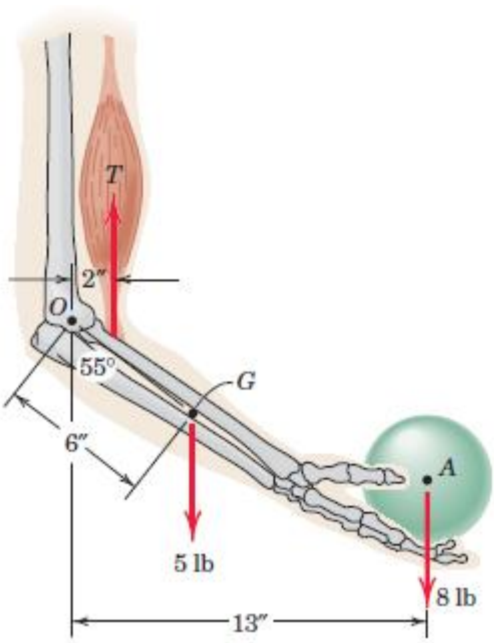
Question 02: Determine the magnitude and direction measured counterclockwise from the positive x axis of the resultant force of the three forces acting on the ring A. It is given that θ = 20 and F_1 = 500 N.



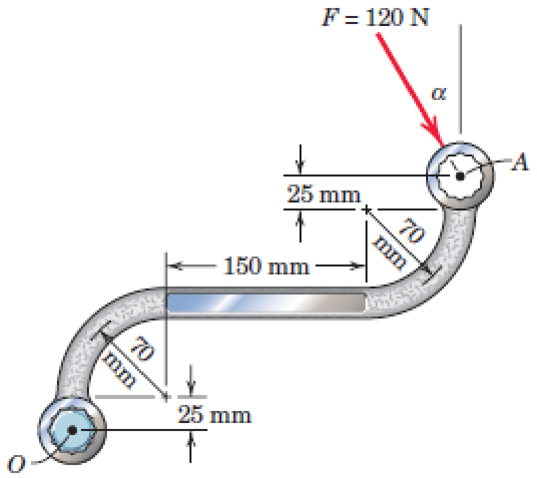
Question 03: A prybar is used to remove a nail as shown. Determine the moment of the 60-lb force about the point O of contact between the prybar and the small support block.



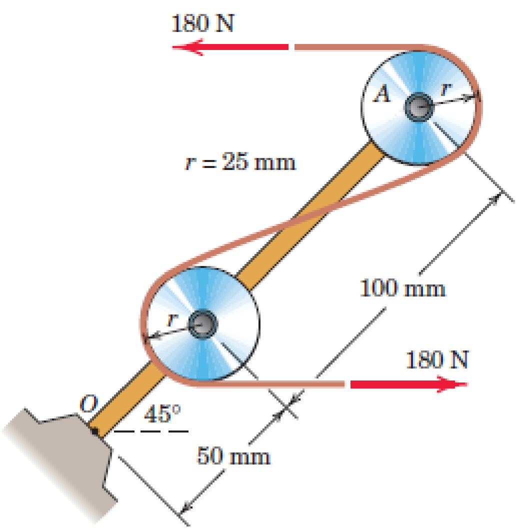
Question 04: Elements of the lower arm are shown in the figure. The weight of the forearm is 5 lb with mass center at G. Determine the combined moment about the elbow pivot O of the weights of the forearm and the sphere. What must the biceps tension force be so that the overall moment about O is zero?



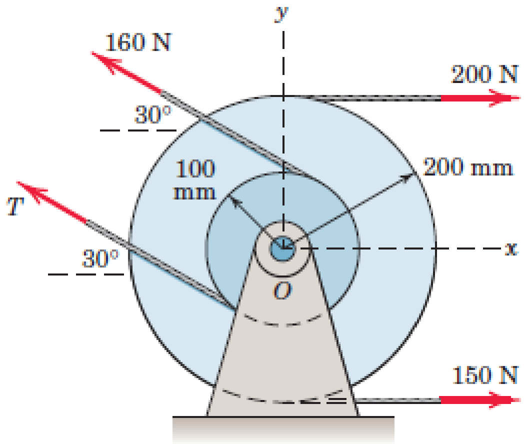
Question 05: The 120-N force is applied as shown to one end of the curved wrench. If calculate the moment of F about the center O of the bolt. Determine the value of which would maximize the moment about O; state the value of this maximum moment.



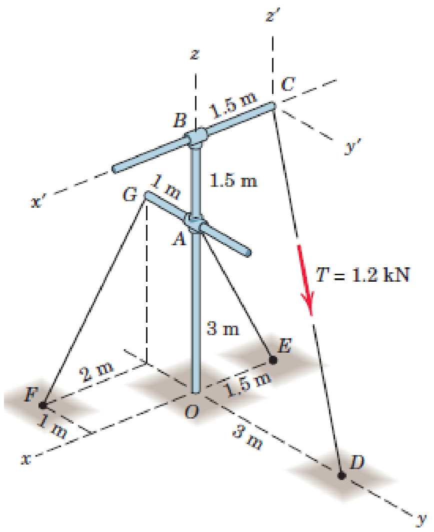
Question 06: The system consisting of the bar OA, two identical pulleys, and a section of thin tape is subjected to the two 180-N tensile forces shown in the figure. Determine the equivalent force–couple system at point O.



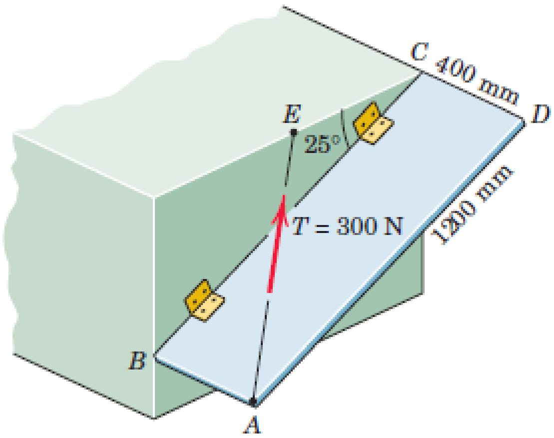
Question 07: Two integral pulleys are subjected to the belt tensions shown. If the resultant R of these forces passes through the center O, determine T and the magnitude of R and the counterclockwise angle it makes with the x-axis.



Question 08: The rigid pole and cross-arm assembly is supported by the three cables shown. A turnbuckle at D is tightened until it induces a tension T in CD of 1.2 kN. Express T as a vector. Does it make any difference in the result which coordinate system is used?



Question 09: The rectangular plate is supported by hinges along its side BC and by the cable AE. If the cable tension is 300 N, determine the projection onto line BC of the force exerted on the plate by the cable. Note that E is the midpoint of the horizontal upper edge of the structural support.



Question 10: Determine the moment about point A and B of each of the three forces acting on the beam.